



System Analysis and Design

CSE 307

Lecture 04

Information Gathering: Interactive Methods



Objectives

- Define system requirements and differentiate between functional and nonfunctional requirements.
- Recognize the value of interactive methods for information gathering.
- Construct interview questions to elicit human information requirements and structure them in a way that is meaningful to users.
- Understand the purpose of stories and why they are useful in systems analysis.
- Understand the concept of JAD and when to use it.
- Write effective questions to survey users about their work.
- Design and administer effective questionnaires.



Interactive Methods to Elicit Human Information Requirements

- Interviewing
- Joint Application Design (JAD)
- Questionnaires



Major Topics

- Interviewing
 - Interview preparation
 - Question types
 - Arranging questions
 - The interview report
- User Stories
- Joint Application Design (JAD)
 - Involvement
 - Location
- Questionnaires
 - Writing questions
 - Using scales
 - Design
 - Administering



Introduction to Requirements Discovery

Requirements discovery includes those techniques to be used by systems analysts to identify or extract system problems and solution requirements from the user community.

Problem analysis is the activity of identifying the problem, understanding the problem (including causes and effects), and understanding any constraints that may limit the solution.



Introduction to Requirements Discovery

A system requirement (also called a business requirement) is a description of the needs and desires for an information system. A requirement may describe functions, features (attributes), and constraints.



Types of Requirements

A functional requirement is a function or feature that must be included in an information system in order to satisfy the business need and be acceptable to the users.

A nonfunctional requirement is a description of the features, characteristics, and attributes of the system as well as any constraints that may limit the boundaries of the proposed solution.

Types of Nonfunctional Requirements

Requirement Type	Explanation
Performance	Performance requirements represent the performance the system is required to exhibit to meet the needs of users. • What is the acceptable throughput rate? • What is the acceptable response time?
Information	Information requirements represent the information that is pertinent to the users in terms of content, timeliness, accuracy, and format. • What are the necessary inputs and outputs? When must they happen? • What is the required data to be stored? • How current must the information be? • What are the interfaces to external systems?
	Economy requirements represent the need for the system to reduce costs or increase profits. • What are the areas of the system where costs must be reduced? • How much should costs be reduced or profits be increased? • What are the budgetary limits? • What is the timetable for development?
Control (and Security)	Control requirements represent the environment in which the system must operate, as well as the type and degree of security that must be provided. • Must access to the system or information be controlled? • What are the privacy requirements? • Does the criticality of the data necessitate the need for special handling (backups, offsite storage, etc.) of the data?

Types of Nonfunctional Requirements (concluded)

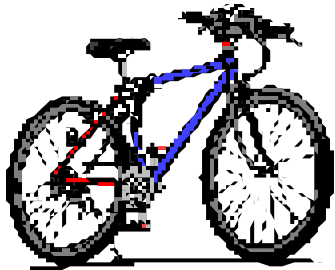
Requirement Type	Explanation
Efficiency	Efficiency requirements represent the systems ability to produce outputs with minimal waste. · Are there duplicate steps in the process that must be eliminated? · Are there ways to reduce waste in the way the system uses it resources?
Service	Service requirements represent needs in order for the system to be reliable, flexible, and expandable. · Who will use the system and where are they located? · Will there be different types of users? · What are the appropriate human factors? · What training devices and training materials are to be included in the system? · What training devices and training materials are to be developed and maintained separately from the system, such as stand- alone computer based training (CBT) programs or databases? · What are the reliability/availability requirements? · How should the system be packaged and distributed? · What documentation is required?

An Ambiguous Requirements Statement

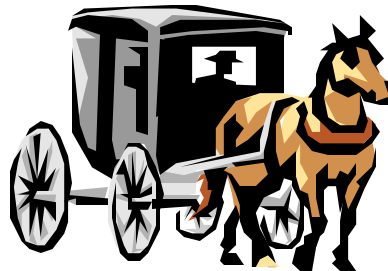
Requirement:

Create a means to transport a single individual from home to place of work.

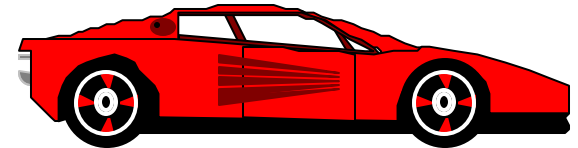
Management
Interpretation



IT
Interpretation



User
Interpretation





Results of Incorrect Requirements

- **The system may cost more than projected.**
- **The system may be delivered later than promised.**
- **The system may not meet the users' expectations and that dissatisfaction may cause them not to use it.**
- **Once in production, the costs of maintaining and enhancing the system may be excessively high.**
- **The system may be unreliable and prone to errors and downtime.**
- **The reputation of the IT staff on the team is tarnished because any failure, regardless of who is at fault, will be perceived as a mistake by the team.**



Criteria to Define System Requirements

- Consistent
- Complete
- Feasible
- Required
- Accurate
- Traceable
- Verifiable



The Process of Requirements Discovery

- Problem discovery and analysis
- Requirements discovery
- Documenting and analyzing requirements
- Requirements management



Requirements Discovery

Fact-finding is the formal process of using research, interviews, questionnaires, sampling, and other techniques to collect information about problems, requirements, and preferences. It is also called information gathering.



Seven Fact-Finding Methods

- Sampling of existing documentation, forms, and databases.
- Research and site visits.
- Observation of the work environment.
- Questionnaires.
- Interviews.
- Prototyping.
- Joint requirements planning (JRP).



Documenting and Analyzing Requirements

A requirements definition document should consist of the following.

- The functions and services the system should provide.
- Nonfunctional requirements including the system's features, characteristics, and attributes.
- The constraints that restrict the development of the system or under which the system must operate.
- Information about other systems the system must interface with.



Sample Requirements Definition Outline

Requirements Definition Report

1. Introduction

1.1 Purpose

1.2 Background

1.3 Scope

1.4 Definitions, Acronyms, and Abbreviations

1.5 References

2. General Project Description

2.1 System Objectives

3. Requirements and Constraints

3.1 Functional Requirements

3.2 Nonfunctional Requirements

4. Conclusion

4.1 Outstanding Issues

Appendix (optional)



Validating Requirements

Requirements validation is an activity that checks the requirements definition document for accuracy, completeness, consistency, and conformance to standards.



Requirements Management

Requirements management is the process of managing change to the requirements.



Interviewing

- Interviewing is an important method for collecting data on human and system information requirements
- Interviews reveal information about:
 - Interviewee opinions
 - Interviewee feelings
 - Goals
 - Key HCI concerns



Interview Preparation

- Reading background material
- Establishing interview objectives
- Deciding whom to interview
- Preparing the interviewee
- Deciding on question types and structure



Question Types

- Open-ended
- Closed



Open-Ended Questions

- Open-ended interview questions allow interviewees to respond how they wish, and to what length they wish
- Open-ended interview questions are appropriate when the analyst is interested in breadth and depth of reply



Advantages of Open-Ended Questions

- Puts the interviewee at ease
- Allows the interviewer to pick up on the interviewee's vocabulary
- Provides richness of detail
- Reveals avenues of further questioning that may have gone untapped



Advantages of Open-Ended Questions (continued)

- Provides more interest for the interviewee
- Allows more spontaneity
- Makes phrasing easier for the interviewer
- Useful if the interviewer is unprepared



Disadvantages of Open-Ended Questions

- May result in too much irrelevant detail
- Possibly losing control of the interview
- May take too much time for the amount of useful information gained
- Potentially seeming that the interviewer is unprepared
- Possibly giving the impression that the interviewer is on a “fishing expedition”



Closed Interview Questions

- Closed interview questions limit the number of possible responses
- Closed interview questions are appropriate for generating precise, reliable data that is easy to analyze
- The methodology is efficient, and it requires little skill for interviewers to administer



Benefits of Closed Interview Questions

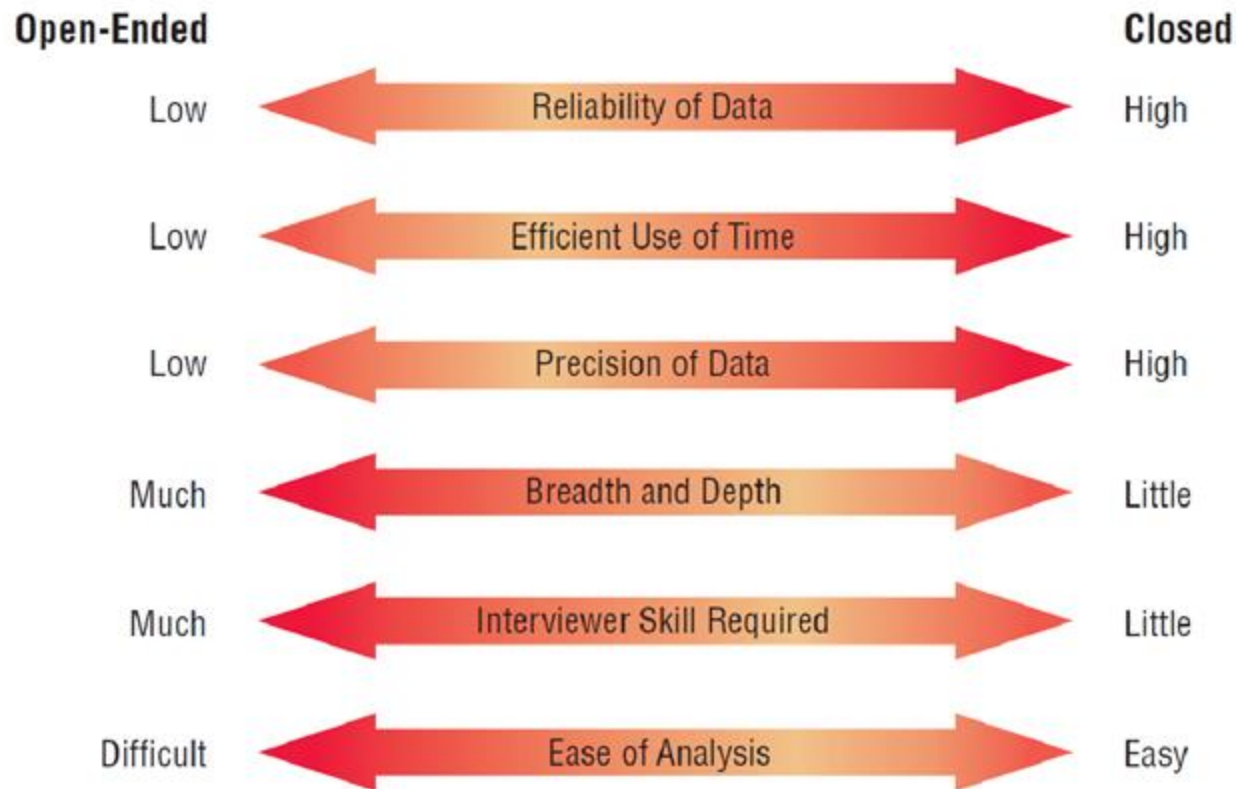
- Saving interview time
- Easily comparing interviews
- Getting to the point
- Keeping control of the interview
- Covering a large area quickly
- Getting to relevant data



Disadvantages of Closed Interview Questions

- Boring for the interviewee
- Failure to obtain rich detailing
- Missing main ideas
- Failing to build relationship between interviewer and interviewee

Attributes of Open-Ended and Closed Questions (Figure 4.5)





Bipolar Questions

- Bipolar questions are those that may be answered with a “yes” or “no” or “agree” or “disagree”
- Bipolar questions should be used cautiously
- A special kind of closed question



Probes

- Probing questions elicit more detail about previous questions
- The purpose of probing questions is:
 - To get more meaning
 - To clarify
 - To draw out and expand on the interviewee's point
- May be either open-ended or closed



Arranging Questions

○Pyramid

- Starting with closed questions and working toward open-ended questions

○Funnel

- Starting with open-ended questions and working toward closed questions

○Diamond

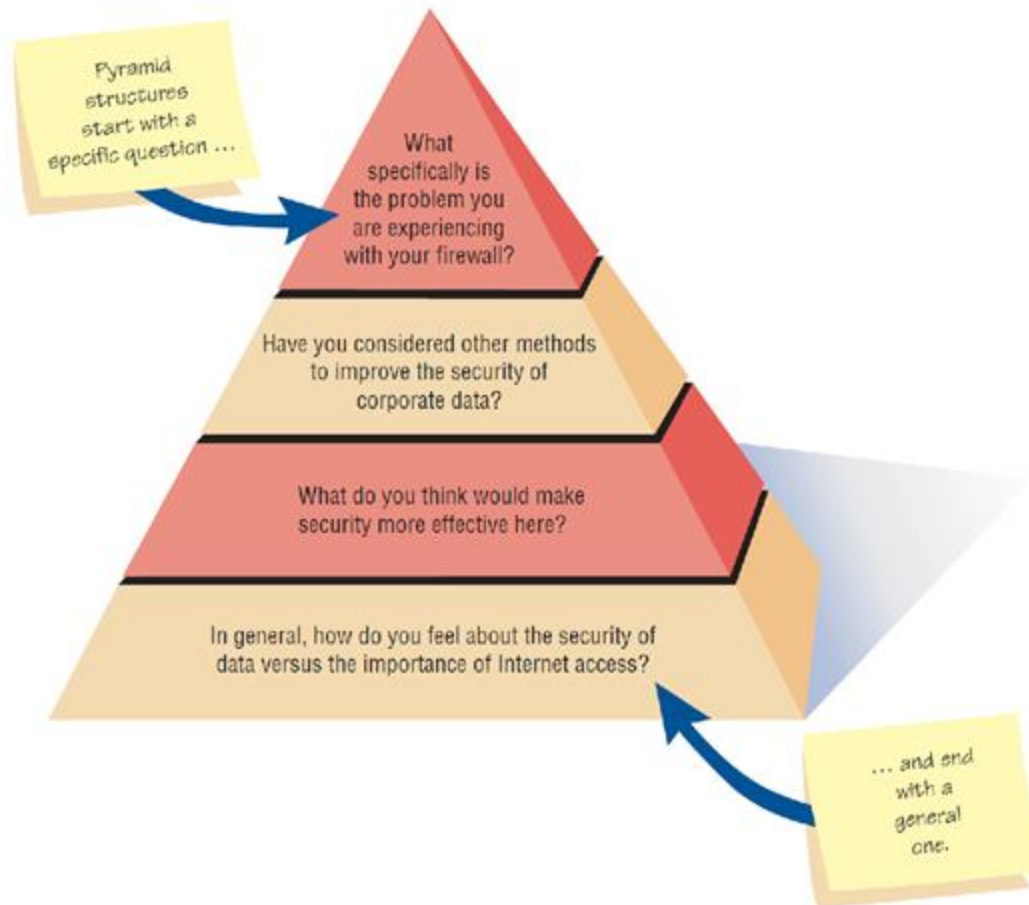
- Starting with closed, moving toward open-ended, and ending with closed questions



Pyramid Structure

- Begins with very detailed, often closed questions
- Expands by allowing open-ended questions and more generalized responses
- Is useful if interviewees need to be warmed up to the topic or seem reluctant to address the topic

Pyramid Structure for Interviewing Goes from Specific to General Questions (Figure 4.7)

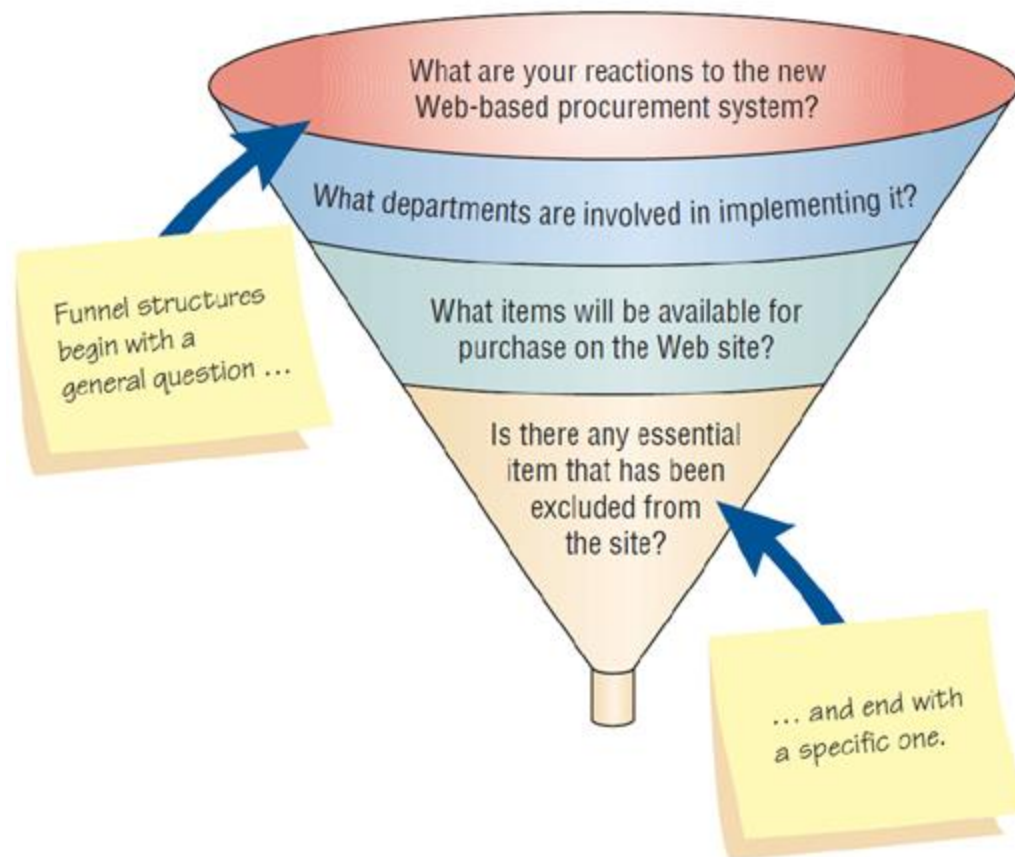




Funnel Structure

- Begins with generalized, open-ended questions
- Concludes by narrowing the possible responses using closed questions
- Provides an easy, nonthreatening way to begin an interview
- Is useful when the interviewee feels emotionally about the topic

Funnel Structure for Interviewing Begins with Broad Questions then Funnels to Specific Questions (Figure 4.8)

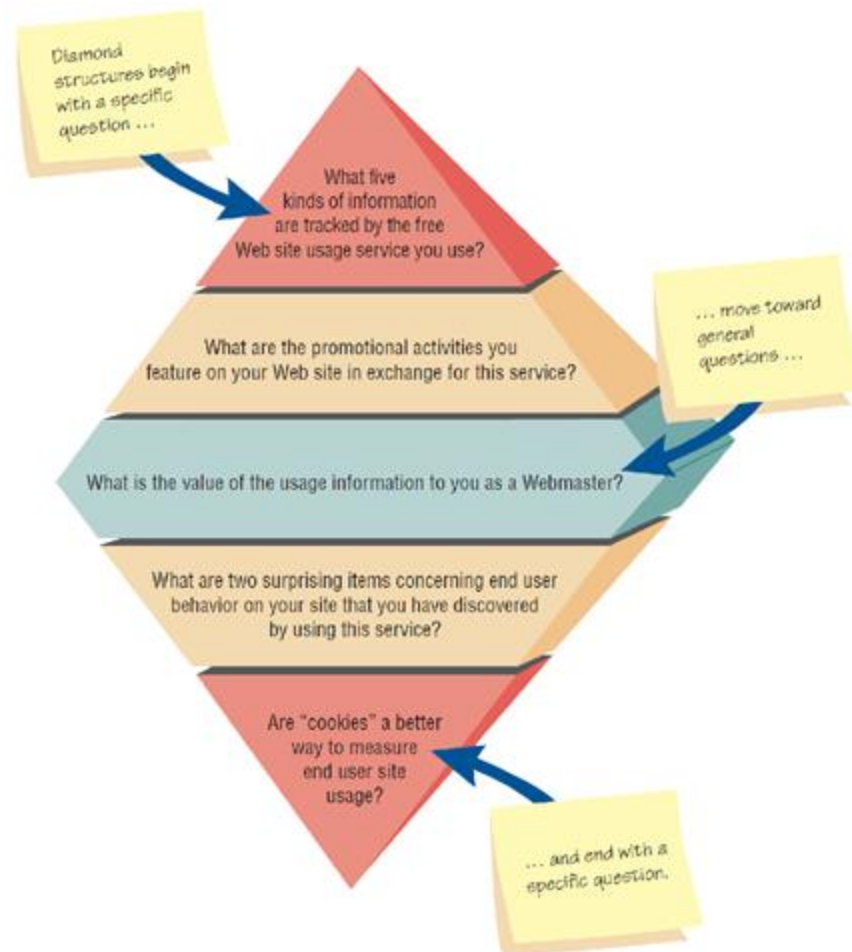




Diamond Structure

- A diamond-shaped structure begins in a very specific way
- Then more general issues are examined
- Concludes with specific questions
- Combines the strength of both the pyramid and funnel structures
- Takes longer than the other structures

Diamond-Shaped Structure for Interviewing Combines the Pyramid and Funnel Structures (Figure 4.9)





Closing the Interview

- Always ask “Is there anything else that you would like to add?”
- Summarize and provide feedback on your impressions
- Ask whom you should talk with next
- Set up any future appointments
- Thank them for their time and shake hands.



Interview Report

- Write as soon as possible after the interview
- Provide an initial summary, then more detail
- Review the report with the respondent



Stories

- Stories originate in the workplace
- Organizational stories are used to relay some kind of information
- When a story is told and retold over time it takes on a mythic quality
- Isolated stories are good when you are looking for facts
- Enduring stories capture all aspects of the organization and are the ones a systems analyst should look for



Purposes for Telling a Story

- There are four purposes for telling a story:
 - Experiential stories describe what the business or industry is like
 - Explanatory stories tell why the organization acted a certain way
 - Validating stories are used to convince people that the organization made the correct decision
 - Prescriptive stories tell the listener how to act
- Systems analysts can use storytelling as a complement to other information gathering methods



Joint Application Design (JAD)

- Joint Application Design (JAD) can replace a series of interviews with the user community
- JAD is a technique that allows the analyst to accomplish requirements analysis and design the user interface with the users in a group setting



Conditions That Support the Use of JAD

- Users are restless and want something new
- The organizational culture supports joint problem-solving behaviors
- Analysts forecast an increase in the number of ideas using JAD
- Personnel may be absent from their jobs for the length of time required



Who Is Involved

- Executive sponsor
- IS analyst
- Users
- Session leader
- Observers
- Scribe



Where to Hold JAD Meetings

○ Offsite

- Comfortable surroundings
- Minimize distractions

○ Attendance

- Schedule when participants can attend
- Agenda
- Orientation meeting



Benefits of JAD

- Time is saved, compared with traditional interviewing
- Rapid development of systems
- Improved user ownership of the system
- Creative idea production is improved



Drawbacks of Using JAD

- JAD requires a large block of time to be available for all session participants
- If preparation or the follow-up report is incomplete, the session may not be successful
- The organizational skills and culture may not be helpful to a JAD session



Questionnaires

Questionnaires are useful in gathering information from key organization members about:

- Attitudes
- Beliefs
- Behaviors
- Characteristics



Planning for the Use of Questionnaires

- Organization members are widely dispersed
- Many members are involved with the project
- Exploratory work is needed
- Problem solving prior to interviews is necessary



Question Types

Questions are designed as either:

- Open-ended
 - Try to anticipate the response you will get
 - Well suited for getting opinions
- Closed
 - Use when all the options may be listed
 - When the options are mutually exclusive

Trade-offs between the Use of Open-Ended and Closed Questions on Questionnaires

(Figure 4.12)





Questionnaire Language

- Simple
- Specific
- Short
- Not patronizing
- Free of bias
- Addressed to those who are knowledgeable
- Technically accurate
- Appropriate for the reading level of the respondent



Measurement Scales

- The two different forms of measurement scales are:
 - Nominal
 - Interval



Nominal Scales

- Nominal scales are used to classify things
- It is the weakest form of measurement
- Data may be totaled

What type of software do you use the most?

1 = Word Processor

2 = Spreadsheet

3 = Database

4 = An Email Program

Interval Scales

- An interval scale is used when the intervals are equal
- There is no absolute zero
- Examples of interval scales include the Fahrenheit or Centigrade scale

How useful is the support given by the Technical Support Group?				
NOT USEFUL			EXTREMELY	
AT ALL			USEFUL	
1	2	3	4	5



Validity and Reliability

- Reliability of scales refers to consistency in response—getting the same results if the same questionnaire was administered again under the same conditions
- Validity is the degree to which the question measures what the analyst intends to measure



Problems with Scales

- Leniency
- Central tendency
- Halo effect



Leniency

○ Caused by easy raters

- Solution is to move the “average” category to the left or right of center



Central Tendency

- Central tendency occurs when respondents rate everything as average
 - Improve by making the differences smaller at the two ends
 - Adjust the strength of the descriptors
 - Create a scale with more points



Halo Effect

- When the impression formed in one question carries into the next question
- Solution is to place one trait and several items on each page



Designing the Questionnaire

- Allow ample white space
- Allow ample space to write or type in responses
- Make it easy for respondents to clearly mark their answers
- Be consistent in style



Order of Questions

- Place most important questions first
- Cluster items of similar content together
- Introduce less controversial questions first



Administering Questionnaires

- Administering questionnaires has two main questions:
 - Who in the organization should receive the questionnaire
 - How should the questionnaire be administered

Ways to Capture Responses When Designing a Web Survey (Figure 4.13)

Name	Appearance	Purpose
One-line text box		Used to obtain a small amount of text and limit the answer to a few words
Scrolling text box		Used to obtain one or more paragraphs of text
Check box	☐	Used to obtain a yes-no answer (e.g., Do you wish to be included on the mailing list?)
Radio button	☐	Used to obtain a yes-no or true-false answer
Drop-down menu		Used to obtain more consistent results (Respondent is able to choose the appropriate answer from a predetermined list [e.g., a list of state abbreviations])
Push button		Most often used for an action (e.g., a respondent pushes a button marked "Submit" or "Clear")



Methods of Administering the Questionnaire

- Convening all concerned respondents together at one time
- Personally administering the questionnaire
- Allowing respondents to self-administer the questionnaire
- Mailing questionnaires
- Administering over the Web or via email



Electronically Submitting Questionnaires

- Reduced costs
- Collecting and storing the results electronically



Summary

- Interviewing
 - Interview preparation
 - Question types
 - Arranging questions
 - The interview report
- Stories
- Joint Application Design (JAD)
 - Involvement and location
- Questionnaires
 - Writing questions
 - Using scales and overcoming problems
 - Design and order
 - Administering and submitting



This work is protected by United States copyright laws and is provided solely for the use of instructors in teaching their courses and assessing student learning. Dissemination or sale of any part of this work (including on the World Wide Web) will destroy the integrity of the work and is not permitted. The work and materials from it should never be made available to students except by instructors using the accompanying text in their classes. All recipients of this work are expected to abide by these restrictions and to honor the intended pedagogical purposes and the needs of other instructors who rely on these materials.

Copyright © 2014 Pearson Education, Inc.
Publishing as Prentice Hall



END OF
Lecture 04